Low-Rank Spectral–Spatial Super-Resolution of Hyperspectral Images Using KAN-Based GAN

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***Abstract*—Hyperspectral image (HSI) super-resolution (SR) must enhance spatial detail while preserving per-pixel spectral signatures. Convolutional and adversarial models often favor perceptual sharpness at the cost of spectral fidelity, whereas purely subspace methods underfit nonlinear interband relations. We present FBD-KAN, a factorized basis decomposition framework trained adversarially, which couples: 1) a low-rank spectral projection/reconstruction path with; 2) Kolmogorov–Arnold network (KAN) layers specialized for smooth yet nonlinear spectral mappings; and 3) lightweight spatial refinements. The generator employs FastKAN-1D layers to project spectra into a compact latent subspace, a residual spectral block to stabilize and enhance interband consistency, and FastKAN-2D with pixel shuffle for spatial upsampling; a bilinear skip stabilizes high-frequency content. Training uses a PatchGAN discriminator and a compound objective [L1, adversarial, and spectral angle mapper (SAM)] to jointly optimize spatial fidelity and spectral accuracy. Across five benchmarks (Salinas, Pavia University, Pavia Centre, Cuprite, CAVE), FBD-KAN attains up to 40.52-dB PSNR with SSIM greater than 0.98 and low SAM while maintaining competitive parameter efficiency. Qualitative spectra further confirm faithful signature reconstruction. Extensive ablation experiments confirm the necessity of each module in the proposed model. Stress testing under Poisson–Gaussian noise shows stable spatial reconstruction but expected spectral sensitivity under photon-limited conditions. Additional interpretability analysis demonstrated that the model learns physically meaningful subspaces and exposes predictable failure modes. Overall, FBD-KAN provides an accurate, interpretable, and computationally lightweight solution for HSI super-resolution, offering strong potential for real-time and onboard remote sensing deployment.**

***Index Terms*—Generative adversarial networks (GANs), hyperspectral super-resolution, Kolmogorov–Arnold networks (KANs), low-rank factorization, spectral–spatial modeling.**

I. Introduction

HYPERSPECTRAL imaging (HSI) has become indispensable in a wide range of remote sensing applications, including environmental monitoring, precision agriculture, mineral exploration, and urban planning. However, the spatial resolution of hyperspectral sensors is often limited due to physical constraints in sensor design, tradeoffs between spectral and spatial resolution, and limitations in onboard hardware. As a result, low-resolution (LR) hyperspectral data frequently fail to capture fine spatial structures critical for accurate downstream analysis. This bottleneck has driven considerable interest in the field of hyperspectral image super-resolution (HSI-SR), where the objective is to reconstruct a high-resolution (HR) hyperspectral cube from its spatially degraded counterpart.

In recent years, generative adversarial networks (GANs) have emerged as a powerful class of models for super-resolution tasks [1]. Unlike traditional convolutional neural networks (CNNs) that minimize pixel-wise loss functions, GANs introduce an adversarial learning paradigm that encourages the generator to produce outputs indistinguishable from real high-resolution images. This framework effectively sharpens the reconstructed images and restores perceptually important high-frequency details, which are often lost in L1- or L2-based reconstructions. GANs have proven especially beneficial in enhancing the realism of RGB images, and their adaptation to hyperspectral data is a natural but challenging progression due to the high spectral dimensionality and information redundancy inherent in HSI [2], [3], [4], [5], [6], [7].

One of the key characteristics of hyperspectral data is its high spectral redundancy, which implies that most of the spectral variance can be captured using a low-dimensional representation. This motivates the use of spectral–spatial factorization, wherein the high-dimensional spectral data are first projected into a low-rank latent space, processed spatially, and subsequently reconstructed back into the original spectral domain. This factorized formulation reduces computational complexity, improves generalization, and better leverages the underlying structure of the data.

While classical linear factorization techniques (e.g., PCA and SVD) are limited in their ability to capture nonlinear interband relationships, recent advances in neural modeling have introduced more expressive alternatives. In this context, Kolmogorov–Arnold networks (KANs) offer a theoretically grounded and functionally rich framework. Rooted in the Kolmogorov–Arnold representation theorem, KANs decompose multivariate continuous functions into compositions of univariate functions and addition operations [8], [9]. This unique construction enables KANs to approximate complex nonlinear mappings with fewer parameters and layers while preserving interpretability and functional smoothness [10], [11]. When deployed in the spectral domain, KANs naturally encode the underlying smooth, yet nonlinear, spectral transitions characteristic of hyperspectral signals.

One core property of HSI is that spectra vary smoothly within materials yet exhibit nonlinear distortions due to

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illumination, atmospheric effects, and subtle compositional mixtures. While low-rank projections (e.g., PCA/SVD) capture redundancy, their linearity limits faithfulness under such nonlinearities. Conversely, deep CNN/Transformer-based SR models can fit complex mappings but may require large depth and data, often tilting the spatial–spectral tradeoff toward perceptual sharpness and away from spectral fidelity.

We address this with KANs, whose construction—compositions of univariate smooth functions and sums—matches the smooth-but-nonlinear character of HSI spectra. In *FBD-KAN*, we: 1) project spectra into a compact latent space with FastKAN-1D layers to impose a low-dimensional spectral manifold; 2) refine the latent spectra via a residual KAN block to stabilize interband relations; and 3) perform spatial upsampling using FastKAN-2D and pixel shuffle, with a bilinear skip for stable high frequencies. A PatchGAN discriminator and a compound loss (L1, adversarial, and SAM) jointly enforce spatial realism and spectral correctness. Experiments on five datasets show consistent gains in PSNR/SSIM with lower SAM, indicating faithful spectral recovery.

Beyond spatial SR, a complementary line of work studies spectral super-resolution (SSR) from RGB, i.e., recovering high-dimensional spectra from trichromatic inputs. Coarse-to-fine CNNs and wavelet-based hybrid asymmetric networks have shown promising SSR performance by injecting spectral priors into RGB-to-HSI mappings [12], [13]. While such SSR methods target spectral recovery from severely underdetermined inputs, our focus is spatial SR of HSIs—recovering high-resolution spatial detail while preserving per-pixel spectra. We briefly discuss SSR to position FBD-KAN within the broader landscape and emphasize that our factorized spectral–spatial design and KAN bias address a different but complementary problem setting.

The key contributions of this work are as follows.

1) We propose FBD-KAN, a spectrally aware SR framework that couples low-rank spectral factorization with KAN layers, aligning the model's inductive bias with smooth, nonlinear HSI spectra.
2) A residual spectral block in latent space consolidates interband consistency; a bilinear skip stabilizes high frequencies during spatial upsampling and eases optimization.
3) A compound objective (L1, adversarial, and SAM) jointly optimizes spatial fidelity and spectral angular accuracy.
4) Across five benchmarks (Salinas, PaviaU, Pavia Centre, Cuprite, and CAVE), FBD-KAN attains strong PSNR/SSIM with consistently lower SAM; qualitative spectra indicate faithful signature reconstruction.

II. Related Work

The field of HSI-SR has witnessed significant progress with the advent of deep learning, primarily driven by the need to recover high-resolution spatial detail while preserving spectral integrity [11], [14], [15], [16]. Earlier HSI-SR methods relied on classical approaches such as sparse representation, interpolation, and coupled dictionary learning. These methods, however, suffered from poor generalization under varying sensor conditions and were computationally expensive.

A. CNN-Based Methods

The early wave of deep learning models for HSI-SR was dominated by CNNs, which exploited local spatial correlations. Yuan et al. [17] proposed a 3D-CNN model that captured spectral–spatial correlations effectively. Li et al. [18] developed a spectral–spatial residual network to further reduce spectral distortion. Xue et al. [19] extended this by integrating sparse low-rank priors into a CNN framework. Dou et al. [6] advanced this by combining 3-D attention with SRGANs. Despite these contributions, standard CNNs face limitations in modeling long-range dependencies, leading to oversmoothing and loss of spectral fidelity.

B. GAN-Based Frameworks

GANs marked a paradigm shift in image super-resolution by introducing adversarial loss to enhance perceptual quality. Zhang et al. [20] introduced RankGAN to capture band-specific features in hyperspectral data. The R2D2-GAN [21] used dual discriminators to better model spatial detail and structural realism. Similarly, Yang et al. [22] employed a hybrid loss combining adversarial, perceptual, and content-based components for SR. Li et al. [23] explored a multiscale GAN with channel attention modules, boosting performance on datasets like CAVE and PaviaU. While these models improved perceptual sharpness, they often increased model complexity and still struggled with spectral artifacts, particularly in underrepresented bands.

C. General Image Super-Resolution

Beyond HSI-specific frameworks, single-image super-resolution (SISR) for natural RGB imagery has advanced from convolutional pyramids and GAN-driven perceptual models to Transformer and physics-inspired architectures. Early CNN-based approaches such as LapSRN [24] achieved stable progressive upsampling, while adversarial models like ESRGAN [5] improved perceptual sharpness through enhanced residual blocks and relativistic discrimination. Transformer-based networks further exploited long-range dependencies, as seen in texture transformer SR (TTSR) [25]. However, when these architectures are applied directly to hyperspectral data (100–200 bands), they exhibit two consistent failure modes: 1) prohibitively high memory and computational cost due to treating each spectral band as an independent RGB channel expansion and 2) loss of spectral identity, since the models are not designed to preserve per-band monotonicity or material-specific absorption peaks.

More recent efficient-attention designs—e.g., ESSAformer [26], which uses spectral-similarity-aware kernels—reduce the computational burden for RGB images but still scale poorly for high-dimensional spectra and do not enforce cross-band coherence. In parallel, physics-inspired SISR models have emerged, such as FMNet [27], which interprets feature propagation via fluid-flow dynamics, and HTI-Net [28], which models residual diffusion using heat-transfer equations. While effective

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for RGB textures, these formulations are incompatible with hyperspectral structure: fluid/heat diffusion implicitly mixes neighboring channels, whereas HSI spectra exhibit sharp, nondiffusive absorption features and require strict band-wise independence to avoid material mixing.

Overall, these general SISR advances underscore the importance of efficient global modeling, yet their assumptions do not generalize to hyperspectral data. This motivates the explicit spectral–spatial factorisation adopted in FBD-KAN, where low-rank spectral projection and KAN-based nonlinear mapping preserve spectral integrity, while lightweight spatial refinement enhances resolution.

D. Spectral SR From RGB

Spectral super-resolution (SSR) aims to reconstruct hyperspectral signatures from RGB observations. Coarse-to-fine CNNs learn RGB-to-HSI mappings by progressively refining spectral detail [12], and wavelet-based hybrid asymmetric networks leverage multiscale transforms to recover high-frequency spectral components from limited color inputs [13]. These approaches address a highly ill-posed spectral inversion problem. In contrast, our work assumes hyperspectral inputs and seeks to enhance their spatial resolution while preserving spectra; thus, SSR and FBD-KAN are complementary: SSR tackles spectral completion from RGB, whereas FBD-KAN performs spatial enhancement under a factorized spectral–spatial inductive bias.

E. Low-Rank and Factorization Techniques

Many recent approaches reduce model complexity and redundancy using low-rank approximations. Dian et al. [29] used tensor low-rank decomposition with joint optimization, while Liu et al. [30] embedded spectral attention into transformers with low-rank priors. Yokoya et al. [31] presented a coupled nonnegative matrix factorization approach that remains influential in model-based SR pipelines. Peng et al. [32] proposed a subspace learning method using group-sparsity factorization. While these techniques offer spectral interpretability and efficient parameterization, they often depend on predefined bases or linear assumptions, limiting expressivity in highly nonlinear scenarios.

F. Kolmogorov–Arnold Networks

KANs provide a principled framework to approximate multivariate functions using univariate transformations and additive compositions, inspired by the Kolmogorov–Arnold representation theorem. This enables the modeling of highly nonlinear spectral functions without deep stacking or overparameterization [8]. Ferdaus et al. [33] proposed KANICE, which integrates KAN layers with interactive convolution elements. Alonso et al. [34] analytically compared KANs with standard neural networks in terms of expressivity and approximation stability. Alesiani et al. [35] presented a geometric interpretation of the Kolmogorov–Arnold theorem in KAN design. These models have demonstrated interpretability, strong generalization, and parameter efficiency—properties desirable for spectral modeling in HSI-SR.

G. Our Contribution

The proposed FBD-KAN introduces a novel, spectrally aware architecture that integrates: 1) low-rank spectral projection and reconstruction using FastKAN 1D layers; 2) spatial refinement using FastKAN 2-D convolutions with pixel shuffle; and 3) a residual spectral mapping block in latent space. A multiterm loss function (pixel-wise, adversarial, and SAM) ensures both perceptual and spectral fidelity. Our framework outperforms previous state-of-the-art approaches in quantitative metrics (PSNR, SSIM, and SAM) across multiple datasets while also offering improved training efficiency and reduced model complexity.

H. Positioning

Prior HSI-SR methods either lean on linear factorization (efficient but limited under nonlinear spectra) or on deep spatial models (expressive but data-hungry and prone to spectral artifacts). Our approach bridges this gap by explicitly factorizing the spectral manifold while endowing the spectral path with a KAN inductive bias—smooth univariate bases composed additively—thereby capturing nonlinear spectral relations with modest depth. In contrast to pure CNN/Transformer SR, adversarial training in FBD-KAN is constrained by an SAM-aware objective, prioritizing spectral correctness.

III. Proposed Methodology

A. Motivation

HSI-SR remains a complex problem due to the inherent tradeoff between spatial detail and spectral fidelity. Deep convolutional networks have shown promise by learning joint spatial-spectral features directly from data. However, many of these models emphasize spatial enhancement, often at the expense of spectral accuracy. Additionally, the increasing depth and parameter count of convolutional architectures raise concerns related to overfitting, computational efficiency, and interpretability. A key property of hyperspectral data is its significant spectral redundancy—many bands are highly correlated, and the intrinsic dimensionality is much lower than the raw band count. This observation suggests a natural pathway: spectral factorization, where the data are projected into a compact, low-dimensional spectral space, processed spatially in this compressed domain, and subsequently reconstructed back. This decouples the spectral and spatial modeling tasks, leading to more efficient and structured learning.

However, classical factorization methods such as principal component analysis (PCA) or low-rank tensor decompositions are inherently linear and struggle to capture the nonlinear spectral variations introduced by factors like illumination changes, atmospheric interference, or subtle material distinctions. This motivates the use of more expressive function approximators capable of modeling nonlinear interband relationships. KANs offer such a framework. Based on the Kolmogorov–Arnold representation theorem, these networks express multivariate functions as compositions of univariate nonlinear transformations and summations. KANs are particularly well-suited to hyperspectral modeling, where smooth nonlinear interactions

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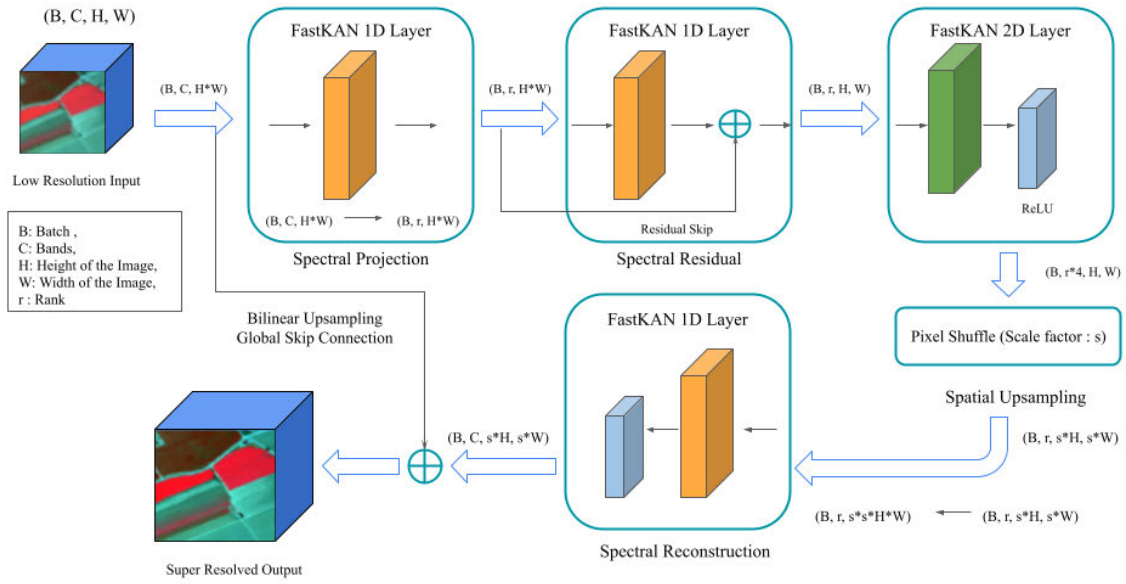


Fig. 1. Network architecture diagram for the generator of the proposed GAN.

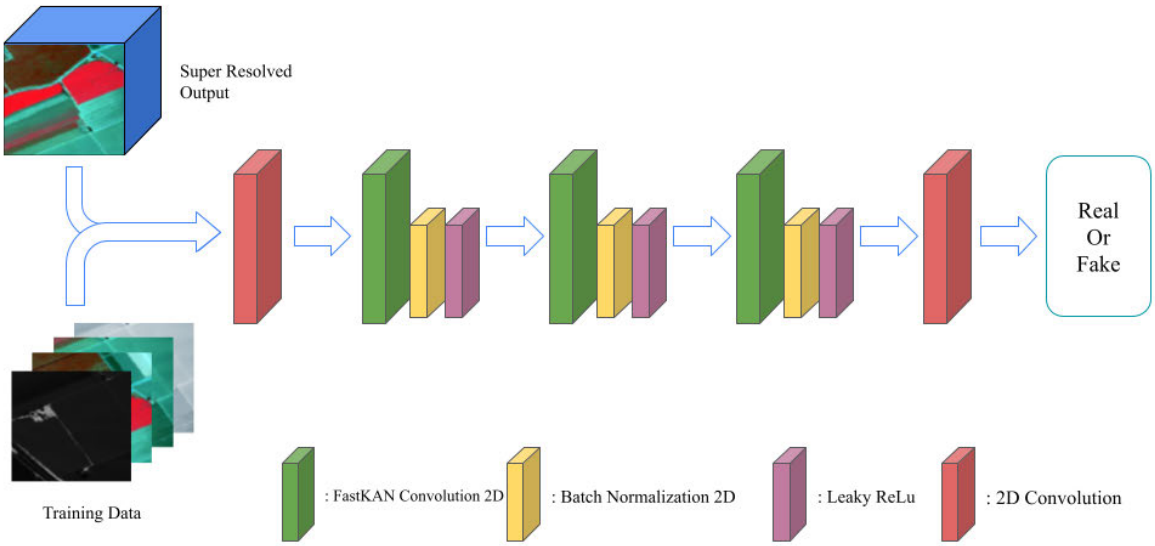


Fig. 2. Network architecture diagram for the discriminator of the proposed GAN.

between bands can be efficiently captured without relying on deep or overparameterized architectures. Their structured nature also enhances model interpretability and stability.

To enable practical deployment of KANs in high-dimensional image tasks, we employ FastKAN—a convolutional implementation of KANs proposed by Drokin [36]. FastKAN extends the KAN formulation by incorporating grid-based basis expansions [e.g., radial basis functions (RBFs)] within learnable convolutional modules. Each FastKAN layer consists of both a linear convolutional path and a nonlinear functional branch, which are fused to produce expressive yet computationally lightweight outputs. This hybrid design allows the network to approximate complex spectral and spatial mappings without excessive depth or redundancy, making it highly suitable for HSI-SR.

Building on these foundations, we propose FBD-KAN, a super-resolution framework that combines spectral factorization, Kolmogorov–Arnold learning, and adversarial training. The model projects hyperspectral input into a low-rank spectral space using FastKAN-1D layers, refines the representation via

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spectral residual blocks, and reconstructs high-resolution outputs using FastKAN-2D layers with pixel shuffle upsampling. A bilinear skip connection ensures structural consistency, while a PatchGAN-style discriminator guides the generator toward photorealistic and spectrally faithful reconstructions.

By separating and explicitly modeling spectral and spatial processes, FBD-KAN provides a robust, interpretable, and computationally efficient solution to HSI-SR. As shown in our experimental results, this architecture consistently outperforms state-of-the-art methods in both spatial reconstruction and spectral preservation across multiple benchmark datasets.

1) Why KAN for Spectra: Let $\mathbf{x} \in \mathbb{R}^C$ be a spectrum. Many materials admit smooth, piecewise-smooth response curves with modest intrinsic dimension. KAN layers realize mappings $f : \mathbb{R}^C \to \mathbb{R}^r$ through shared banks of smooth univariate basis functions and linear mixing. This provides: 1) sample-efficient approximation of smooth nonlinearities; 2) stable gradients due to smooth bases; and 3) interpretability via basis activations. In FBD-KAN, we use FastKAN implementations that fuse basis expansions with convolutions for efficiency.

2) Why KAN Rather Than MLPs, Spline Bases, or Kernel Regressors: Hyperspectral spectra are *smooth but nonlinear* functions of wavelength. ReLU MLPs approximate such functions via piecewise linear partitions and typically require depth/width growth to achieve smoothness and stability; this increases parameters and can harm spectral fidelity due to sharp activation transitions. In contrast, KAN layers implement shared banks of *smooth univariate basis functions* with additive mixing, aligning with the Kolmogorov–Arnold representation [8], [34], [37]. This yields: 1) sample-efficient approximation of smooth nonlinearities; 2) better-conditioned gradients (no piecewise kinks); and 3) interpretable basis responses per band. Classical spline expansions require manual knot/width selection and do not share bases across bands, whereas KAN shares a learnable basis bank and learns the mixing end-to-end. Nonparametric kernel regressors provide strong local fits but scale at least quadratically with samples and do not amortize parameters across pixels, limiting deployability in HSI-SR. Empirically (Section IV-D), replacing KAN with CNN/MLP surrogates degrades PSNR and increases SAM at similar parameter budgets, supporting the inductive-bias advantage of KAN for spectra.

B. Mathematical Formulation and Network Architecture

1) Spectral–Spatial Factorization: Let $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$ be the LR HSI. We seek a compact spectral representation of rank $r \ll C$. Classical linear projections write

$$\mathbf{Z} \approx \mathbf{W}_s \mathbf{X}_{(1)} \in \mathbb{R}^{r\times(H\cdot W)} \tag{1}$$

with $\mathbf{W}_s \in \mathbb{R}^{r\times C}$. In FBD-KAN, this projection is *realized by a nonlinear FastKAN-1D layer*, which better captures smooth but nonlinear spectral relations while maintaining parameter efficiency.

2) KANs via FastKAN: A FastKAN-1D layer maps a spectrum $\mathbf{x} \in \mathbb{R}^C$ to $\mathbf{z} \in \mathbb{R}^r$ by expanding each input channel with a bank of smooth univariate bases and then linearly mixing

$$\Phi_c(\mathbf{x}_c) = \left[\phi_{c,1}(x_c), \ldots, \phi_{c,M}(x_c)\right] \in \mathbb{R}^M \tag{2}$$

$$\Phi(\mathbf{x}) = \left[\Phi_1(\mathbf{x}_1) \parallel \cdots \parallel \Phi_C(\mathbf{x}_C)\right] \in \mathbb{R}^{\mathrm{CM}} \tag{3}$$

$$\mathbf{z} = \mathbf{W}\Phi(\mathbf{x}) + \mathbf{b}, \quad \mathbf{W} \in \mathbb{R}^{r\times(\mathrm{CM})}. \tag{4}$$

Here, $\phi_{c,m}$ is the smooth grid-aligned basis (e.g., B-splines/RBFs). FastKAN-2D extends this idea to feature maps $\mathbf{U} \in \mathbb{R}^{C\times H\times W}$ by: 1) applying per-channel basis expansions pointwise and 2) aggregating with learnable $k \times k$ convolutions. These FastKAN layers used were originally implemented in [37]. In practice, we use a fused implementation that: 1) computes a linear convolutional path and 2) a basis-expansion path, whose outputs are concatenated and linearly mixed, yielding expressive yet lightweight layers.

3) Factorized Basis Decomposition and Roles of 1-D/2-D KAN: FBD-KAN explicitly decouples spectral and spatial processing. Let $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$. We first apply a *spectral* projector implemented by FastKAN-1D

$$\mathbf{Z} = \mathrm{FastKAN1D}_{\mathrm{proj}}\left(\mathrm{reshape}(\mathbf{X})\right) \in \mathbb{R}^{r\times(HW)}, \quad r \ll C \tag{5}$$

and then reshape $\mathbf{Z} \to \mathbb{R}^{r\times H\times W}$ and refine spectra in a *residual* FastKAN-1D block. This 1-D branch encodes smooth yet nonlinear interband relations with modest parameters and depth, aligning with typical HSI spectral manifolds.

Spatial enhancement is handled by FastKAN-2D layers that aggregate local context with learnable $k \times k$ mixing while retaining a smooth functional path via basis expansions. Producing $s^2 r$ channels followed by PixelShuffle, the 2-D KAN learns to place high-frequency content at correct spatial locations while the spectral residual path preserves per-pixel spectral integrity.

a) Why 1-D KAN Versus 2D KAN (and Versus CNN): The 1-D KAN focuses on per-pixel spectra and implements smooth univariate basis banks with linear mixing, providing sample-efficient nonlinear approximation of spectral curves. The 2-D KAN extends this idea to feature maps and fuses it with convolutional aggregation, enabling spatial detail recovery without deep stacks. Compared to standard CNNs (which rely solely on linear filters + pointwise nonlinearities), KAN paths incorporate explicit smooth basis expansions, yielding stable gradients and improved spectral faithfulness with fewer layers. This division of labor—1-D for spectra and 2-D for space—operationalizes the factorized basis decomposition, which is visually represented in Fig. 1.

4) Serial Versus Parallel Coupling of Low Rank and KAN: Let $\mathcal{P}_{\mathrm{LR}}$ denote the low-rank projector and $\mathcal{F}_{\mathrm{KAN}}$ denote the nonlinear spectral mapper. A *parallel* design applies both directly on X and fuses the outputs

$$Y_{\mathrm{par}} = g\left(\mathcal{P}_{\mathrm{LR}}(X) \parallel \mathcal{F}_{\mathrm{KAN}}(X)\right)$$

where $g(\cdot)$ mixes features. This introduces: 1) *identifiability overlap*—both branches can explain the same low-frequency content; 2) *gradient interference*—the easy-to-optimize linear path steals supervision, starving the nonlinear path; and 3) higher compute due to duplicate encodings. Our *serial* design

$$X \xrightarrow{\mathcal{P}_{\mathrm{LR}}} Z \xrightarrow{\text{res. KAN}} Z_r \xrightarrow{\text{2DKAN + PixelShuffle}} Y$$

first *conditions* spectra onto a compact manifold and then applies KAN where nonlinearity is actually needed, followed

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TABLE I
SUMMARY OF HYPERSPECTRAL DATASETS USED

Dataset	Spatial Resolution	Bands	Sensor	Scene Type
Salinas	512×217	204	AVIRIS	Agricultural fields
Pavia University	610×340	103	ROSIS	Urban environment
Pavia Centre	1096×715	102	ROSIS	Urban city centre
Cuprite	350×350	224	AVIRIS	Mineralogical mapping of arid terrain
CAVE	512×512	31	Synthetic	Indoor lab scenes with objects

by spatial detail placement. This ordering reduces redundancy, improves conditioning of $\mathcal{F}_{\text{KAN}}$ (narrower dynamic range, fewer modes), and lowers FLOPs because the spatial stage operates on $r \ll C$ channels. In Section IV-D, we include a *parallel* variant to quantify the effect.

5) Generator Pipeline: It is given as follows.

a) Spectral Projection: Reshape $\mathbf{X}$ to $\mathbf{X}' \in \mathbb{R}^{C\times N}$, and $N = H \cdot W$. Apply FastKAN-1D

$$\mathbf{Z}_p = \text{FastKAN1D}_{\text{proj}}\left(\mathbf{X}'\right) \in \mathbb{R}^{r\times N} \tag{6}$$

and then, reshape $\mathbf{Z}_p \to \mathbb{R}^{r\times H\times W}$.

b) Residual Spectral Refinement:

$$\mathbf{Z}_r = \mathbf{Z}_p + \text{FastKAN1D}_{\text{res}}\left(\mathbf{Z}_p\right). \tag{7}$$

c) Spatial Upsampling: Apply a FastKAN-2D producing $s^2 r$ channels (for upscale factor s), followed by PixelShuffle

$$\mathbf{U} = \text{FastKAN2D}\left(\mathbf{Z}_r\right) \in \mathbb{R}^{(s^2 r)\times H\times W} \tag{8}$$

$$\mathbf{Z}_{\text{HR}} = \text{PixelShuffle}_s\left(\mathbf{U}\right) \in \mathbb{R}^{r\times \text{sH}\times \text{sW}}. \tag{9}$$

PixelShuffle rearranges channels to spatial locations: for indices (c, i, j)

$$\left[\text{PixelShuffle}_s\left(\mathbf{U}\right)\right]_{c,\, si+a,\, s\ j+b} = \mathbf{U}_{c\cdot s^2+a\cdot s+b,\, i,\, j} \quad a, b \in \{0, \ldots, s-1\}. \tag{10}$$

d) Spectral Reconstruction:

$$\widehat{\mathbf{Y}} = \text{FastKAN1D}_{\text{recon}}\left(\mathbf{Z}_{\text{HR}}\right) \in \mathbb{R}^{C\times \text{sH}\times \text{sW}}. \tag{11}$$

e) Bilinear Skip:

$$\mathbf{Y}_{\text{final}} = \text{Clip}\left(\widehat{\mathbf{Y}} + \text{Upsample}_{\text{bilinear}}\left(\mathbf{X}\right),\, 0,\, 1\right). \tag{12}$$

6) Discriminator: D_ϕ is a PatchGAN with FastKAN-2D blocks and BatchNorm; it outputs a patch map $\mathbf{D} \in \mathbb{R}^{1\times(sH)/\rho\times(sW)/\rho}$ (stride factor ρ). The discriminator flow is represented Fig. 2.

7) Objective: The generator is optimized with a three-term objective combining pixel fidelity, adversarial realism, and spectral angular consistency

$$\mathcal{L}_G = \lambda_{\text{pix}} \left\|\mathbf{Y} - \mathbf{Y}_{\text{gt}}\right\|_1 + \lambda_{\text{adv}}\, \mathcal{L}_{\text{adv}} + \lambda_{\text{sam}}\, \mathcal{L}_{\text{SAM}} \tag{13}$$

where $\mathcal{L}_{\text{adv}} = -1/(|\Omega|) \sum_{p\in\Omega} \log D_\phi(\mathbf{Y})_p$ is the standard generator loss over PatchGAN outputs (set Ω of patch locations), while

$$\mathcal{L}_{\text{SAM}} = \frac{1}{sHsW} \sum_i \arccos\left(\frac{\langle \mathbf{y}_i, \hat{\mathbf{y}}_i\rangle}{\|\mathbf{y}_i\|_2\ \|\hat{\mathbf{y}}_i\|_2 + \varepsilon}\right) \tag{14}$$

averages the per-pixel spectral angle between the reconstruction $\mathbf{y}_i$ and ground truth $\hat{\mathbf{y}}_i$. In our implementation, $\mathcal{L}_{\text{SAM}}$ is aggregated over the mini-batch and used as a scalar penalty to stabilize training. The discriminator uses the usual binary cross-entropy on real/fake patches with logits

$$\mathcal{L}_D = \tfrac{1}{2}\left(\text{BCE}\left(D_\phi\left(\mathbf{Y}_{\text{gt}}\right), \mathbf{1}\right) + \text{BCE}\left(D_\phi\left(\text{stopgrad}\left(\mathbf{Y}\right)\right), \mathbf{0}\right)\right) \tag{15}$$

where stopgrad$(\cdot)$ denotes that the generator output is detached when updating D_ϕ.

IV. EXPERIMENTS AND RESULTS

A. Datasets and Preprocessing

To evaluate the generalization capability of the proposed FBD-KAN architecture, we conduct experiments on five benchmark hyperspectral datasets: Salinas [38], Pavia University [38], Pavia Centre [38], Cuprite [38], and CAVE [39]. These datasets span different spatial resolutions, spectral characteristics, and acquisition environments, offering a comprehensive benchmark for hyperspectral super-resolution, as shown in Table I.

To simulate LR inputs, we adopt a patch-based degradation framework. From each dataset, fixed-size patches of spatial dimension 72×72 and full spectral depth C are extracted using a nonoverlapping sliding window. These high-resolution patches are then spatially downsampled using bicubic interpolation by factors of $2\times$, $3\times$, and $4\times$ to generate LR counterparts. The resulting LR and high-resolution patch pairs serve as inputs and targets, respectively, for supervised training.

For the CAVE dataset, which consists of indoor object-centric scenes captured under controlled illumination, we perform additional preprocessing to ensure compatibility with other datasets. First, images are spectrally normalized to [0, 1] and spatially cropped to a consistent field of view where objects are centered. To enhance training diversity, horizontal and vertical flips, as well as 90° rotations, are applied during runtime augmentation.

All hyperspectral bands affected by sensor noise or water absorption have been removed in preprocessing. Each spectral channel is normalized independently based on global minimum and maximum values across the dataset to ensure consistent spectral scaling.

B. Training Setup

We train with 72×72 LR patches and an upscale factor $s = 2$ (so HR is $2H \times 2W$). We use Adam optimizers

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TABLE II
QUANTITATIVE COMPARISON OF STATE-OF-THE-ART MODELS WITH THE PROPOSED METHOD ACROSS MULTIPLE DATASETS AT SCALE FACTOR ×2

Scale	Model	Salinas			PaviaU			Pavia Centre			CAVE			Cuprite		
		PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM
×2	Bicubic	30.11	0.910	0.089	30.72	0.895	0.098	31.06	0.880	0.103	30.43	0.900	0.092	29.81	0.824	0.110
	3D-FCNN [16]	35.98	0.931	0.075	34.62	0.918	0.083	33.21	0.905	0.090	36.32	0.924	0.078	33.27	0.888	0.063
	3D-SRCNN [40]	36.11	0.933	0.073	34.90	0.918	0.085	33.87	0.906	0.096	36.91	0.922	0.078	33.80	0.901	0.069
	HSI-GAN [41]	37.65	0.944	0.066	35.41	0.927	0.077	34.98	0.917	0.087	38.32	0.936	0.070	34.01	0.900	0.070
	SCFN [42]	38.12	0.951	0.059	35.86	0.934	0.071	35.63	0.925	0.081	39.05	0.942	0.065	34.12	0.901	0.064
	LE-GAN [43]	38.91	0.958	0.053	36.24	0.941	0.065	36.01	0.932	0.075	39.88	0.948	0.060	35.02	0.923	0.069
	FBD-KAN (Ours)	**40.52**	**0.987**	**0.017**	**40.12**	**0.981**	**0.027**	**39.99**	**0.980**	**0.026**	**40.01**	**0.981**	**0.018**	**38.19**	**0.980**	**0.027**

TABLE III
QUANTITATIVE COMPARISON OF STATE-OF-THE-ART MODELS WITH THE PROPOSED METHOD ACROSS MULTIPLE DATASETS AT SCALE FACTOR ×3

Scale	Model	Salinas			PaviaU			Pavia Centre			CAVE			Cuprite		
		PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM
×3	Bicubic	28.48	0.892	0.098	28.01	0.876	0.108	28.42	0.860	0.115	28.72	0.885	0.100	27.85	0.852	0.114
	3D-FCNN [16]	34.07	0.912	0.084	32.94	0.894	0.093	32.01	0.880	0.101	34.62	0.900	0.088	32.50	0.873	0.099
	3D-SRCNN [40]	34.02	0.921	0.091	32.78	0.905	0.103	31.66	0.892	0.118	34.85	0.913	0.100	30.52	0.878	0.125
	HSI-GAN [41]	35.48	0.935	0.085	33.41	0.919	0.095	33.12	0.912	0.101	36.21	0.928	0.091	31.91	0.889	0.111
	SCFN [42]	35.96	0.942	0.078	33.87	0.926	0.091	33.75	0.918	0.096	37.02	0.934	0.084	32.32	0.891	0.102
	LE-GAN [43]	36.79	0.951	0.071	34.26	0.935	0.083	34.14	0.926	0.089	37.88	0.939	0.079	33.02	0.898	0.099
	FBD-KAN (Ours)	**37.68**	**0.978**	**0.013**	**37.23**	**0.980**	**0.08**	**37.12**	**0.979**	**0.024**	**37.96**	**0.976**	**0.015**	**36.95**	**0.975**	**0.021**

[betas = $(0.9, 0.999)$] with an initial learning rate of 1×10^{-4} for both G and D, and StepLR schedulers with step size 40 and decay $\gamma = 0.5$. The default training length is 100 epochs. The PatchGAN discriminator outputs a patch map and is trained with *BCEWithLogitsLoss*; labels are 1 for real and 0 for fake. For each iteration, D is updated on real HR patches and on generator outputs detached from the graph; then, G is updated to: 1) minimize L1 reconstruction error; 2) fool D (adversarial term with target label 1); and 3) minimize the mini-batch averaged spectral angle mapper (SAM).

We set the loss weights to $(\lambda_{\text{pix}}, \lambda_{\text{adv}}, \lambda_{\text{sam}}) = (10.0, 1.0, 0.1)$ and use $\varepsilon = 10^{-8}$ for numerical stability in SAM. Training logs are printed every 50 iterations. Pixel values are scaled to $[0, 1]$; water absorption/noisy bands are removed in preprocessing. Unless stated otherwise, metrics (PSNR/SSIM/SAM) are computed per band and averaged spatially and spectrally.

All these experiments were carried out on a local system with an Intel i7 processor, an Intel Iris Xe graphics card, 16-GB RAM, PyTorch 2.9, and python 3.12.

C. Comparative Study

To assess the effectiveness of our proposed FBD-KAN architecture, we compare its performance against several state-of-the-art HSI-SR models, including 3D-FCNN [16], 3D-SRCNN [40], HSI-GAN [41], SCFN [42], and LE-GAN [43]. Bicubic interpolation is used as a baseline. The evaluation is conducted across five standard hyperspectral datasets: Salinas, PaviaU, Pavia Centre, CAVE, and Cuprite, and reported in terms of PSNR, SSIM, and SAM at scale factors ×2, ×3, and ×4.

As shown in Table II, FBD-KAN achieves the highest PSNR and SSIM while maintaining the lowest SAM across all datasets for the ×2 scale. Notably, on the Salinas dataset, it surpasses various state-of-the-art baselines in PSNR and SAM values, reflecting superior spectral fidelity. Similar improvements are observed on urban datasets such as PaviaU and Pavia Centre, where FBD-KAN maintains consistent spectral alignment and spatial detail reconstruction.

At the ×3 scale (Table III), although the reconstruction task becomes more challenging, FBD-KAN continues to outperform all competing methods. It consistently achieves over 37-dB PSNR on all datasets and maintains SSIM values above 0.976. The SAM values remain markedly lower, indicating that our method preserves spectral signatures even under increased upscaling complexity.

For the most challenging ×4 scale (Table IV), where most models show significant degradation in both spatial and spectral accuracy, FBD-KAN maintains strong generalization. It yields an average gain of around 1.2–1.5 dB in PSNR over other baselines, with consistent SSIM above 0.94 and SAM values substantially lower (e.g., 0.018 on Salinas). This robustness underscores the strength of our frequency-biased decomposition strategy and the effectiveness of KAN-based modeling.

Overall, the experimental results validate that FBD-KAN achieves state-of-the-art performance in HSI-SR tasks across diverse datasets and varying degradation scales. Its consistent edge across all metrics and datasets demonstrates not only high spatial fidelity but also strong preservation of spectral integrity. The qualitative results of above experiments can be seen in Fig. 3.

D. Ablation Study and Pareto Analaysis

To assess the necessity of each architectural component in FBD-KAN, we conduct a comprehensive ablation across: 1) the low-rank spectral projector; 2) the KAN-based nonlinear spectral mapping; 3) the spatial refinement pathway; 4) skip-connection design; and 5) loss terms. In addition to the canonical variants (e.g., KAN→CNN replacement, skip

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TABLE IV

QUANTITATIVE COMPARISON OF STATE-OF-THE-ART MODELS WITH THE PROPOSED METHOD ACROSS MULTIPLE DATASETS AT SCALE FACTOR ×4

Scale	Model	Salinas			PaviaU			Pavia Centre			CAVE			Cuprite		
		PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM
×4	Bicubic	25.92	0.872	0.108	25.61	0.851	0.117	28.95	0.839	0.124	25.47	0.864	0.112	24.78	0.811	0.127
	3D-FCNN [16]	32.18	0.891	0.096	30.82	0.871	0.106	29.89	0.857	0.113	32.56	0.882	0.100	29.21	0.850	0.113
	3D-SRCNN [40]	31.48	0.901	0.109	30.65	0.882	0.119	29.97	0.861	0.131	32.61	0.893	0.112	28.13	0.842	0.139
	HSI-GAN [41]	32.82	0.917	0.098	31.44	0.897	0.109	31.02	0.886	0.117	33.89	0.911	0.101	29.30	0.865	0.127
	SCFN [42]	33.26	0.924	0.092	31.88	0.904	0.104	31.67	0.892	0.112	34.71	0.919	0.096	29.98	0.832	0.119
	LE-GAN [43]	34.03	0.935	0.086	32.36	0.914	0.099	32.14	0.903	0.107	35.52	0.925	0.090	30.02	0.881	0.115
	FBD-KAN (Ours)	**35.23**	**0.955**	**0.018**	**35.89**	**0.939**	**0.029**	**35.99**	**0.940**	**0.028**	**35.89**	**0.958**	**0.023**	**34.54**	**0.935**	**0.023**

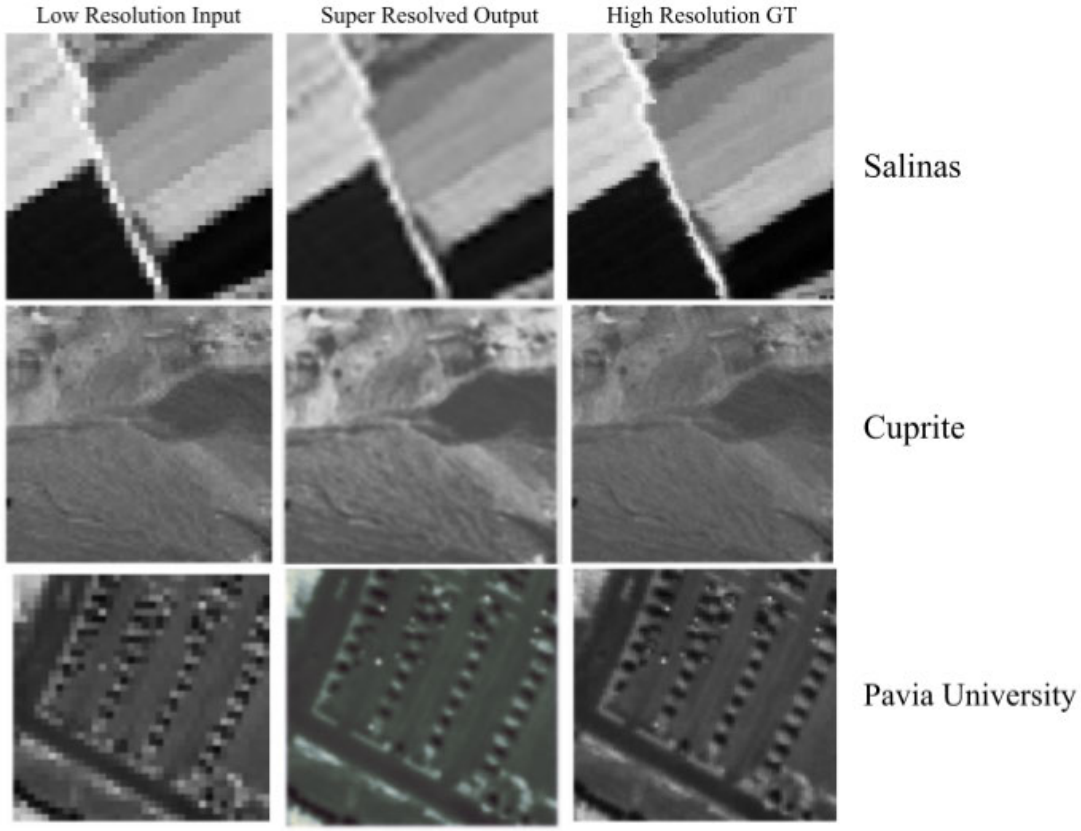


Fig. 3. Qualitative result of the proposed method on different datasets with scale factor 2. The images show simulated low-resolution input, super-resolved output, and the ground-truth images.

TABLE V

ABLATION ON SALINAS AT ×3. METRICS ARE MEAN OVER BANDS (HIGHER PSNR/SSIM AND LOWER SAM). #PARAMS AND FLOPS ARE CONV-EQUIVALENT PER-IMAGE FORWARD PASS

Variant	PSNR	SSIM	SAM	Params (M)	GFLOPs
Full FBD-KAN (Ours)	**37.68**	**0.978**	**0.013**	0.44	2.56
MLP baseline	24.03	0.654	0.210	0.36	1.66
w/o Bilinear skip	25.03	0.723	1.022	0.44	1.85
Bicubic skip	30.90	0.771	0.030	0.44	1.15
L_{pix} only	28.12	0.641	0.034	0.40	1.63
$L_{\text{pix}}+L_{\text{mse}}$	32.35	0.921	0.029	0.51	1.24
w/o Spectral clipping	29.07	0.763	0.028	0.39	1.42
w/o Low-Rank projector	22.08	0.632	0.020	0.42	1.34
w/o Spatial refinement	7.72	0.469	0.028	0.39	1.46
Parallel coupling	6.64	0.428	0.825	13.9	57.79

alternatives, and loss reweighting), we introduce three study cases: 1) removing the low-rank projector; 2) removing spatial refinement; and 3) parallel coupling between the low-rank and KAN branches.

Table V reports PSNR/SSIM/SAM along with parameter count and FLOPs (per 72 × 72 LR patch at ×3 upscaling) to enable complexity–performance comparison.

1) Impact of Low-Rank Spectral Projection: Eliminating the FastKAN-1D projector (*w/o low-rank projector*) degrades PSNR from 37.68 to 22.08 dB despite keeping nearly the same parameter budget, demonstrating that directly processing the full C-dimensional spectrum overwhelms the subsequent nonlinear mapping. This confirms that the learned low-rank subspace is not a heuristic compression, but a functionally necessary spectral manifold that improves conditioning, reduces redundancy, and isolates the physically meaningful endmember mixtures.

2) Necessity of the KAN Nonlinear Spectral Mapper: Replacing all KAN layers with standard convolutions yields a

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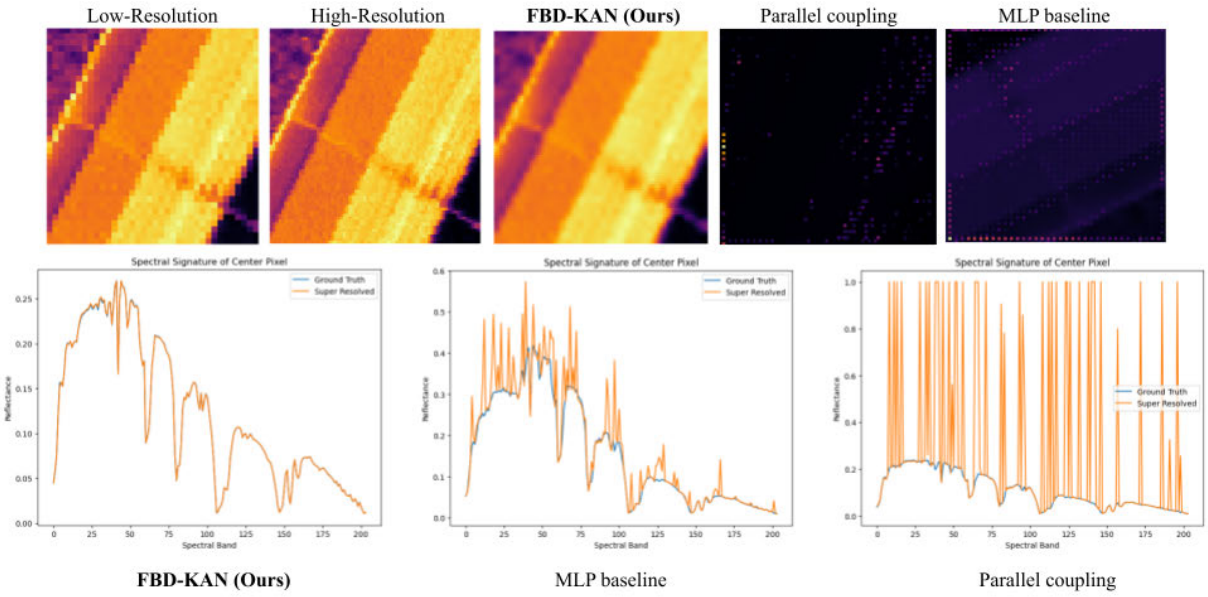


Fig. 4. Visual results of key ablation experiments done with the Salinas dataset at a scale factor of 3. The ablations addressed are the MLP baseline of the proposed model and the parallel coupling of the low-rank and KAN branches. The top row shows the visuals of patches, and the bottom row shows the spectral signatures of the center pixel of images reconstructed in different experiments.

sharper degradation in the *MLP baseline* (24.03 dB), which collapses spatial structure and produces oscillatory spectral signatures. The visualization of degradation can be seen in Fig. 4. Together with the learned KAN radial basis responses, these results validate that KAN delivers the "smooth yet nonlinear" functional mappings required to capture complex interband relationships that are not well modeled by linear convolutions or shallow MLPs.

3) Role of Spatial Refinement: Replacing the FastKAN-2D+PixelShuffle branch with a single bilinear upsampling layer (*w/o spatial refinement*) causes catastrophic failure (7.72 dB: SAM = 0.028), indicating that purely spectral processing is insufficient. The spatial refinement branch is therefore indispensable for restoring high-frequency geometric detail after spectral projection.

4) Skip-Connection Design and Information Lifting: Removing the bilinear skip severely harms all metrics (25.03 dB: SAM = 1.022), revealing that multiresolution identity paths stabilize reconstruction and prevent oversmoothing. Substituting them with bicubic interpolation partially recovers PSNR (30.90 dB) but remains far below the full model, confirming that learnable, band-aligned HR lifting is crucial.

5) Influence of Loss Formulation: Using only the L_{pix} term degrades spectral fidelity (SAM = 0.034), and adding MSE improves it only marginally. The full objective (pixel + SAM) yields the lowest SAM (0.013), reflecting that angular losses are essential to preserve high-dimensional spectral geometry. This aligns with established hyperspectral reconstruction theory, where angular similarity is more correlated with material identity than L1/L2 norms.

6) Effect of Spectral Clipping: Omitting spectral clipping increases SAM from 0.013 to 0.028 (refer to Tables III and V), showing that FastKAN-induced overshoot must be mitigated to maintain reflectance realism and avoid negative or physically implausible spectra.

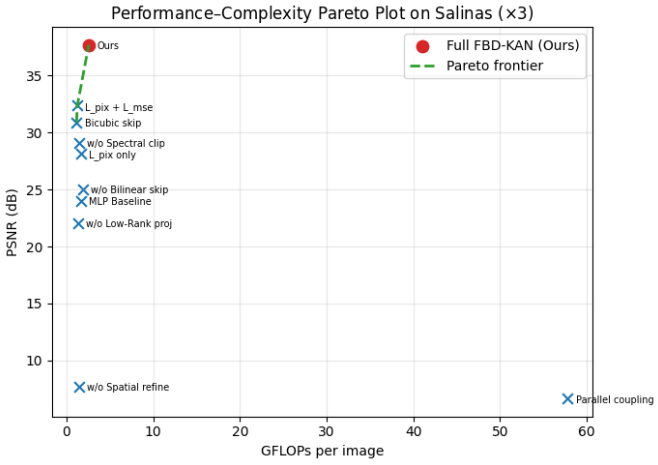


Fig. 5. Graphical illustration for Pareto analysis representing performances of the proposed model (red) and all the variants against computational complexity (FLOPs). The Salinas datasets with a scale factor of 3.

7) Serial Versus Parallel Coupling: The *parallel coupling* variant—where low-rank and KAN branches operate independently on the input and fuse via a 1 × 1 mixer—performs extremely poorly (6.64 dB: SAM = 0.825) while consuming ≈58 GFLOPs. The quantitative results in Table V and qualitative results shown in Fig. 4 confirm that the proposed serial pipeline (low rank → KAN → spatial refine) is not only more accurate but also far more efficient. Parallel paths amplify noise across all channels, lack spectral disentanglement, and collapse when learning long-range nonlinear interactions.

8) Complexity–Performance Tradeoff (Pareto Analysis): Fig. 5 presents a Pareto plot of PSNR versus FLOPs. A configuration is Pareto-optimal if no other variant achieves both higher accuracy and lower complexity. The full FBD-KAN lies on the Pareto frontier; it delivers the best spectral and spatial fidelity while operating at only 2.56 GFLOPs. Notably, the

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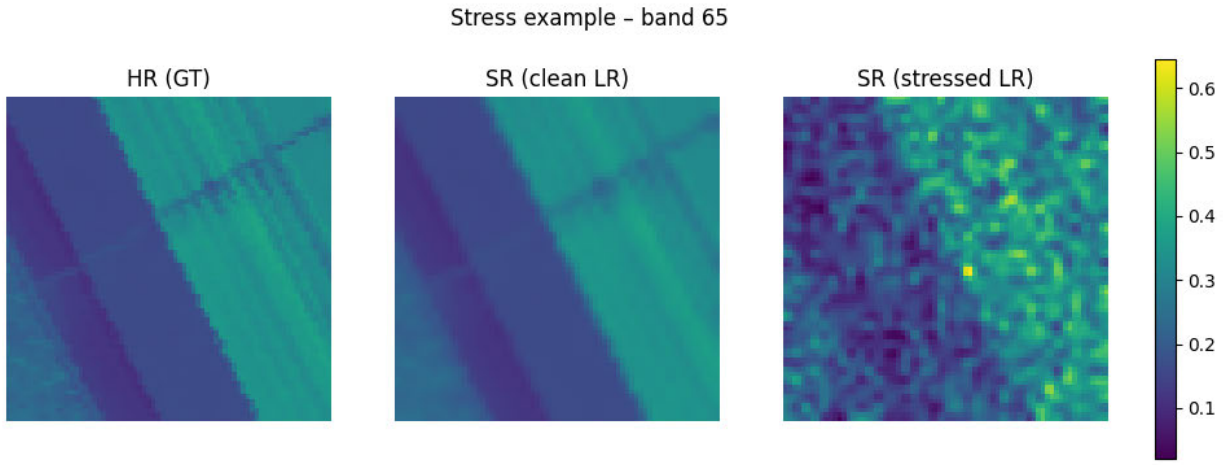


Fig. 6. Stress-test example for band 65. FBD-KAN reconstructs clean LR inputs faithfully, preserving geometry and material boundaries. Under Poisson–Gaussian corruption, the reconstruction becomes unstable, exhibiting amplified noise and disrupted spectral contrast.

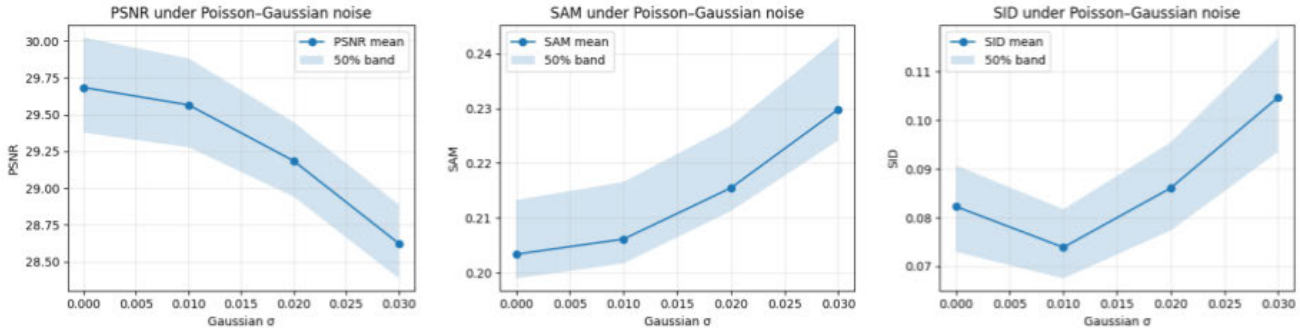


Fig. 7. Performance degradation of FBD-KAN under increasing Poisson–Gaussian noise. Mean PSNR monotonically decreases, while SAM and SID increase, indicating reduced spatial and spectral fidelity. The shaded 50% quantile bands reveal low variance for PSNR/SSIM but widening uncertainty for SAM/SID, highlighting greater sensitivity of spectral metrics to signal-dependent noise.

TABLE VI

ROBUSTNESS OF FBD-KAN UNDER POISSON–GAUSSIAN NOISE. METRICS ARE AVERAGED OVER THE TEST SET. QUANTILES CORRESPOND TO 25%, 50%, AND 75% CONFIDENCE LEVELS. HIGHER PSNR/SSIM AND LOWER SAM/SID INDICATE BETTER SPECTRAL FIDELITY. THESE EXPERIMENTS WERE CARRIED OUT ON THE SALINAS DATASET WITH A SCALE FACTOR OF 2

Metric	Mean (PG Noise)	Quantiles (25 / 50 / 75)
PSNR (dB)	27.31	27.03 / 27.17 / 27.61
SSIM	0.5010	0.4785 / 0.5130 / 0.5259
SAM	0.2655	0.2590 / 0.2692 / 0.2785
SID	0.12667	0.1169 / 0.1225 / 0.1395

parallel coupling design is strictly dominated—it uses over 20× higher FLOPs with far worse fidelity. These results show that the proposed serial low rank→KAN→spatial pipeline is not only accurate but also computationally principled, achieving favorable tradeoffs in the accuracy–complexity plane.

E. Analysis Under Realistic Perturbations

We evaluate the robustness of FBD-KAN under perturbations that emulate noise processes commonly encountered in operational hyperspectral systems (e.g., photon-limited imaging and sensor readout distortion). Figs. 6 and 7 summarize the degradation characteristics, while Table VI reports mean and quantile-based uncertainty estimates.

1) Sensitivity to Poisson–Gaussian Noise: Poisson–Gaussian corruption induces severe spectral–spatial distortion due to simultaneous signal-dependent photon noise and additive readout noise. Under this perturbation, FBD-KAN exhibits a substantial reduction in spatial fidelity; the mean PSNR drops to **27.31 dB**, with a tight quantile spread (**27.03/27.17/27.61 dB**), suggesting that spatial degradation is consistent in the used dataset of Salinas. The corresponding SSIM values follow a similar pattern (mean **0.501**), indicating a uniform attenuation of structural contrast.

In contrast, spectral metrics show disproportionately high sensitivity. SAM increases to a mean of **0.2655**, and SID, which is given by the following equation:

$$\mathrm{SID}(\mathbf{p},\mathbf{q}) = \sum_{i=1}^{C} \left(p_i \ln \frac{p_i}{q_i} + q_i \ln \frac{q_i}{p_i} \right) \tag{16}$$

rises to **0.1267**, both indicating significant perturbation to cross-band relationships. The quantile bands widen more noticeably for SAM/SID compared to PSNR/SSIM, implying that spectral degradation is highly nonuniform across pixels

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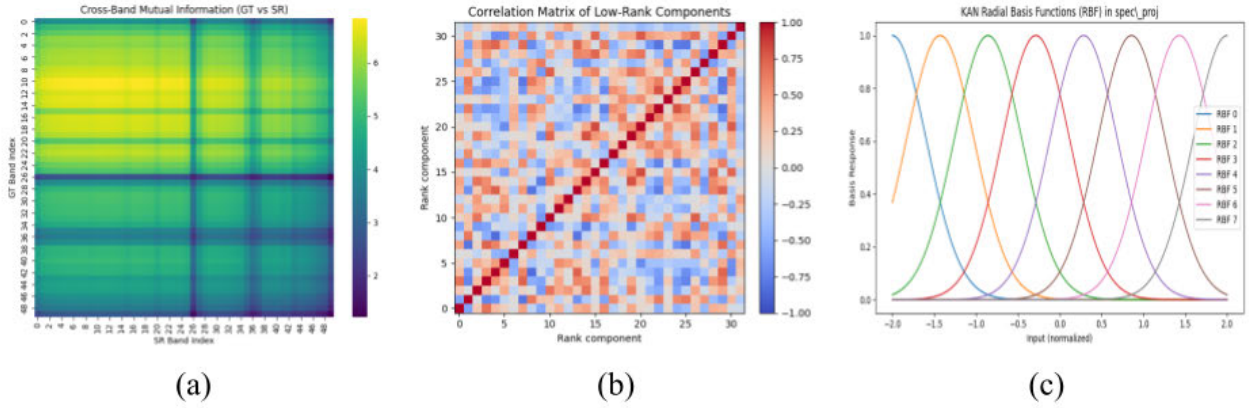


Fig. 8. Interpretability analysis of the proposed FBD-KAN model. (a) Cross-band mutual information (GT versus SR), showing strong diagonal dominance and coherent off-diagonal structure, indicating that the model preserves interband spectral dependencies. (b) Correlation matrix of the learned low-rank components, demonstrating that the FastKAN projector produces largely decorrelated spectral factors. (c) RBF responses of the KAN activation, illustrating smooth, overlapping basis functions that enable nonlinear yet controlled spectral mappings.

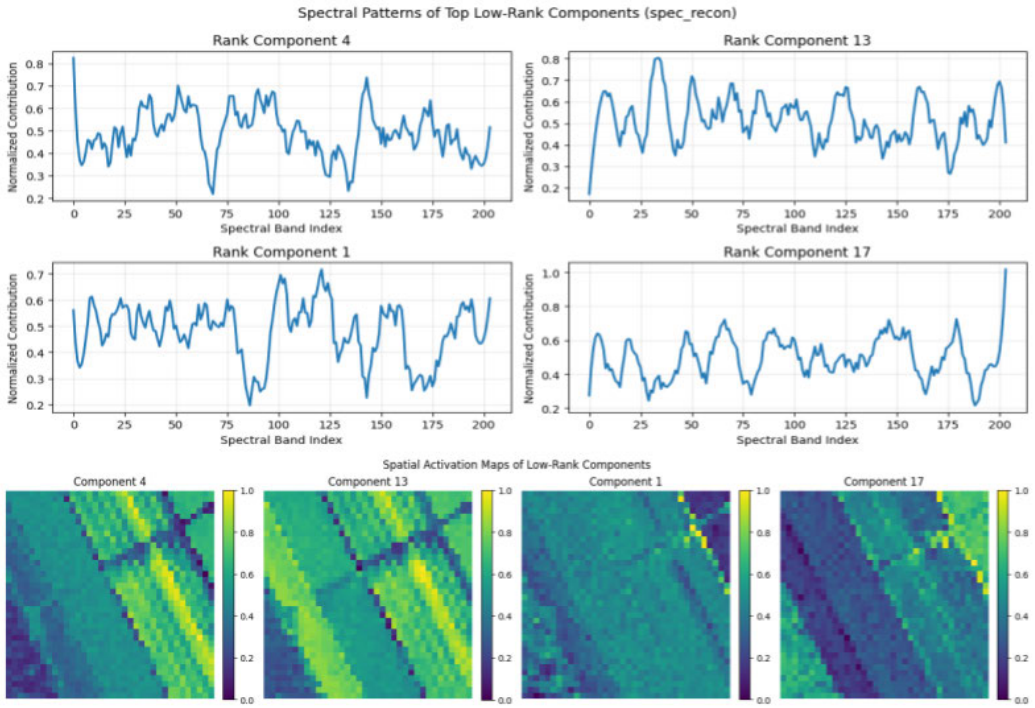


Fig. 9. Learned low-rank spectral components. (Top) Spectral signatures of the four most active components extracted by the low-rank projector. (Bottom) Spatial activation maps for the same components. The pairing of smooth spectral curves with localized spatial responses suggests that the low-rank representation decomposes the scene into meaningful material-aligned subspaces.

and materials. This behavior is consistent with the fact that signal-dependent noise disproportionately disrupts the low-rank spectral structure on which the model relies.

2) Visual Degradation Patterns: The qualitative examples in Fig. 6 demonstrate that noise-dominated LR inputs yield reconstructions contaminated by high-frequency artifacts and unstable spectral contrast. Although coarse geometry is still partially preserved, material boundaries and subtle gradients deteriorate sharply. These observations align with the quantitative rise in SAM/SID rather than PSNR/SSIM, confirming that spectral coherence breaks earlier than spatial structure.

3) Quantile Behavior and Uncertainty Interpretation: The stress curves in Fig. 7 reveal distinct uncertainty trends. Spatial metrics (PSNR and SSIM) retain narrow quantile bands, indicating relatively uniform degradation across scenes. In contrast, the quantile bands for SAM and SID widen rapidly as noise intensity increases, signaling heterogeneous spectral vulnerability: low-reflectance surfaces, shadowed regions, and mixed-material pixels are disproportionately affected. The divergence between median and upper quantiles suggests that certain spectral profiles suffer catastrophic breakdown even when average performance appears moderate.

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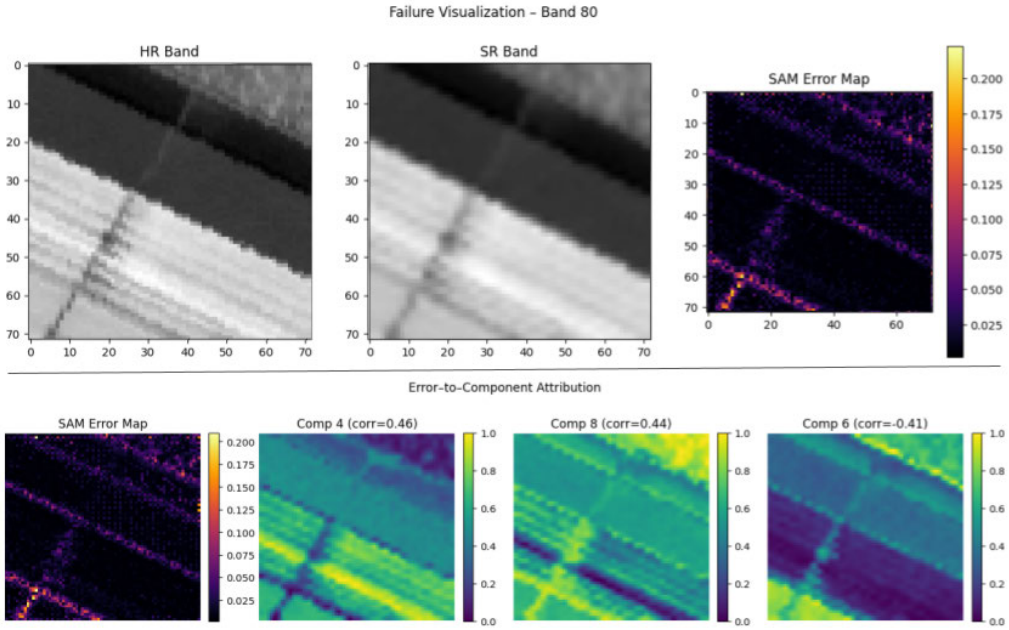


Fig. 10. Failure case and componentwise error attribution for band 80. (Top row) Ground truth (HR), reconstructed output (SR), and the per-pixel SAM error map, highlighting regions with strong spectral distortion (shadows and material boundaries). (Bottom row) SAM error map alongside the three low-rank components that exhibit the highest correlation with the error. These components (Comp 4, Comp 8, and Comp 6) capture fine-grained spectral variations that the model struggles to reconstruct consistently, indicating which subspace directions contribute most to the failure.

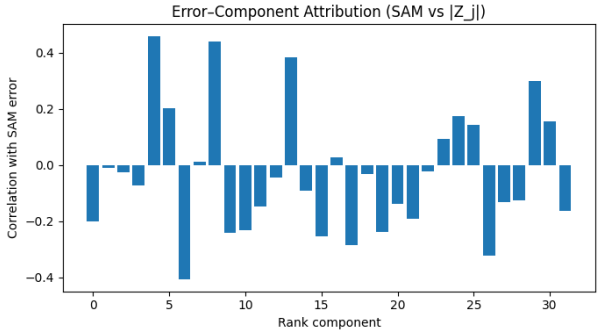


Fig. 11. Correlation between per-pixel spectral error (SAM) and the absolute activation of each low-rank component. Only a small subset of components shows strong positive or negative correlation, indicating that specific spectral–spatial subspace directions dominate the model's failure modes. Components with high correlation correspond to edge- or texture-sensitive bases, revealing where spectral mismatch originates within the low-rank representation.

4) Robustness Interpretation: Overall, FBD-KAN demonstrates stable spatial robustness but pronounced spectral sensitivity under photon-limited conditions. The model retains coarse geometric structure reliably, yet the low-rank spectral projector and nonlinear KAN mapping are susceptible to disruption when cross-band statistical coherence is weakened by Poisson–Gaussian noise. This behavior reflects underlying physics-driven constraints rather than model-specific instability: realistic HSI degradation disproportionately corrupts spectral relationships, making SAM/SID essential indicators of early failure.

F. Model Interpretability and Failure Analysis

To demonstrate that FBD-KAN learns meaningful spectral–spatial representations and displays interpretability, we analyze the learned low-rank subspaces, KAN nonlinear responses, cross-band statistical consistency, and attribution-guided failure modes. Together, these results reveal how each module contributes to the reconstruction process and where the model is susceptible to error.

1) Cross-Band Mutual Information and Spectral Consistency: Fig. 8(a) presents the cross-band mutual information (MI) between ground truth and reconstructed spectra. High MI along the diagonal indicates strong band-wise fidelity, while coherent off-diagonal structure reflects physically meaningful spectral mixing. Crucially, no artificial block patterns or spurious correlations are introduced. This establishes that FBD-KAN preserves interband relationships and does not distort the underlying spectral geometry.

2) Component Independence and Diversity: The correlation matrix shown in Fig. 8(b) illustrates that rank components exhibit low off-diagonal correlation, confirming that the learned subspace directions are diverse and minimally redundant. No pathological collinearity is observed across the 32-dimensional subspace, supporting the interpretation of the low-rank projector as a structured basis extractor.

3) Smooth Nonlinear Mapping via KAN Basis Functions: Fig. 8(c) shows the RBFs underlying the KAN spectral layers. The responses are smooth, continuous, and locally supported, in contrast to the piecewise linear activations of MLPs or CNNs. Their overlapping structure enables fine-grained spectral interpolation while preserving global smoothness. This empirically confirms that KAN layers realize smooth yet nonlinear spectral mappings.

4) Low-Rank Spectral Subspace Structure: Fig. 9 visualizes the dominant spectral bases extracted by the low-rank projector along with their spatial activation maps. The spectral curves (top panel) exhibit smooth, slowly varying patterns across

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wavelengths, resembling material endmembers or mixed-subspace signatures. High-energy components align with the characteristic reflectance of vegetation, soil, and shadowed regions in the Salinas scene. The corresponding spatial maps (bottom panel) show that different rank components respond to distinct regions—textured vegetation, soil background, and long homogeneous strips—indicating that the low-rank projection decomposes the scene into meaningful material-aligned subspaces. This confirms that the low-rank branch performs more than dimensionality reduction; it constructs a physically consistent spectral manifold.

5) Representative Failure Modes: Fig. 10 provides a joint visualization of HR, SR, SAM error distribution, and the spatial maps of the top error-correlated rank components. Spectral errors concentrate in thin structures (crop boundaries), mixed-material transitions, and shadowed regions. The corresponding low-rank components exhibit spatial activation in precisely these locations, providing a clear attribution pathway from failure regions to internal model representations. This demonstrates that failure modes are not arbitrary but arise from predictable interactions between rank components and local scene complexity.

6) Error-to-Component Attribution: Fig. 11 plots the correlation between per-pixel SAM error and the magnitude of each low-rank component. Only a few components exhibit strong positive or negative correlation, primarily those associated with high-frequency or edge-sensitive spectral patterns. This identifies the specific subspace directions that are most vulnerable to spectral mismatch.

V. Conclusion

This work presented FBD-KAN, a factorized spectral–spatial super-resolution framework that combines low-rank spectral projection, Kolmogorov–Arnold nonlinear mapping, and lightweight spatial refinement within an adversarial generator–discriminator setup. Evaluated across five benchmark HSI datasets and multiple scaling factors ($\times 2$, $\times 3$, and $\times 4$), FBD-KAN delivers state-of-the-art spatial fidelity and spectral accuracy, consistently outperforming CNN- and GAN-based baselines in PSNR, SSIM, and SAM.

A key practical contribution is the model's computational efficiency. All experiments—training and inference—were performed on a standard local machine, demonstrating that the factorized KAN design is deployable without dedicated accelerators, making it suitable for low-resource research settings and field platforms.

Development analyses highlighted several stability factors: KAN layers require bounded activations and spectral clipping to avoid reflectance overshoot; the bilinear skip connection is essential for preserving high-frequency structure; and ablations confirm that each component—low-rank projector, KAN spectral block, and serial factorized coupling—contributes critically to reconstruction quality and FLOP efficiency. Interpretability studies further showed that the model learns meaningful spectral subspaces, smooth KAN basis responses, and component-specific failure modes.

Perturbation experiments revealed that while spatial reconstruction remains stable, spectral geometry is more sensitive to photon-limited noise, indicating value in integrating physics-aware priors or robust denoising mechanisms.

Overall, FBD-KAN demonstrates that low-rank spectral manifolds, KAN-based nonlinear inductive biases, and lightweight spatial refinement form an effective, interpretable, and resource-efficient solution for hyperspectral super-resolution.

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